RTL9 (Q8NET4): A Domesticated Retroviral Gag Gene Functioning as a Microglial Lysosomal Antifungal Effector

Gene: RTL9 (Retrotransposon Gag-like protein 9) **Synonyms:** RGAG1, KIAA1318, SIRH10, MART9, Tumor antigen BJ-HCC-23 **UniProt:** Q8NET4 · **NCBI Gene:** 57529 · **Locus:** Xq23 · **Organism:** *Homo sapiens*

Gene Identity Verification

Before presenting the research findings, the mandatory identity check was performed. The target gene symbol **RTL9** and its synonyms (RGAG1, KIAA1318, SIRH10, Mart9) all map consistently to **UniProt Q8NET4**, a 1388-amino-acid human protein located on the X chromosome (Xq23). The UniProt protein description — "Retrotransposon Gag-like protein 9 / Retrotransposon gag domain-containing protein 1" — is fully concordant with the key domain annotation (**PEG10-RTL1_C-capsid**, PF30901 / IPR063300; and the RTL9-specific signature IPR042919). The scientific literature on RTL9/SIRH10 in the eutherian RTL/SIRH gene family (primarily from the Ishino/Kaneko-Ishino group) describes the same gene and the same retroviral-Gag-derived architecture. **Identity is confirmed; the research below pertains to the correct gene.**

One important caveat for readers mining databases: the symbol **RGAG1** also appears in several clinical/genomic papers (BCOR fusion sarcomas, Xq deletion syndromes, testicular cancer methylation studies, UK10K lipid GWAS). These describe the *genomic locus* RGAG1/RTL9 as a bystander or fusion partner and do **not** characterize the biochemical function of the protein. The functional annotation below is drawn from the RTL/SIRH functional literature, not these incidental genomic reports.

Summary

RTL9 (RGAG1/SIRH10/Mart9; UniProt **Q8NET4**) is a **domesticated retroviral Gag gene** — one of eleven eutherian-specific **Retrotransposon Gag-like (RTL) / sushi-ichi-related retrotransposon homolog (SIRH)** genes that arose when a Ty3/Gypsy ("sushi-ichi") LTR-retrotransposon was exapted into the mammalian genome. The human protein is 1388 residues (~144 kDa), encoded by a single-copy, X-linked gene at Xq23. Its only recognizable structured module is a **C-terminal retroviral Gag/capsid-like domain** (Pfam PF30901 / InterPro IPR063300), mapped to approximately residues 1239–1305; the large N-terminal region is poorly characterized and substantially disordered.

The **primary function of RTL9 is innate antifungal immunity in the brain**. Experimental evidence from RTL9-mCherry knock-in mice shows that the RTL9 protein localizes to **microglial lysosomes**, where it degrades **zymosan** — a fungal cell-wall β-glucan/mannan preparation that serves as a model fungal pathogen-associated molecular pattern (PAMP). Critically, in knockout mice engineered to express an **RTL9ΔC** protein lacking the C-terminal retroviral Gag-like region, this zymosan-degrading activity was abolished, demonstrating that the **Gag domain is essential** for function (PMID: 37834332).

RTL9 is therefore not a classical enzyme, transporter, or signaling receptor. Instead, it is the **antifungal arm of a three-member microglial innate-immune module** unique to the eutherian brain: **RTL5/SIRH8 → viruses (double-stranded RNA), RTL6/SIRH3 → bacteria (lipopolysaccharide), and RTL9/SIRH10 → fungi (zymosan)**. Whereas RTL5 and RTL6 act extracellularly as secreted PAMP-binding granules, RTL9 is distinctive in acting **intracellularly within the lysosome** to clear phagocytosed fungal material. The human–mouse orthologs are strongly conserved across the functional Gag module (~78% amino-acid identity), supporting conservation of this antifungal role in humans.

Key Findings

Finding 1 — Primary function: lysosomal antifungal defense in brain microglia

The single most informative study of RTL9 function is **Shiura et al., 2023** (*Retrovirus-Derived RTL9 Gene..., Int. J. Mol. Sci.*; PMID: 37834332). Using **RTL9-mCherry knock-in mice**, the authors demonstrated that the RTL9 protein is expressed in **microglia** — the brain's resident

macrophages — and that it **localizes specifically to lysosomes**. When fungal **zymosan** (a particulate β-1,3-glucan/mannan preparation derived from the *Saccharomyces cerevisiae* cell wall and a canonical fungal PAMP) was introduced into the brain, it was phagocytosed by microglia and trafficked to lysosomes, where RTL9 mediated its **degradation**. In the authors' own words: *"Here, we demonstrate that RTL9 plays an important role, degrading zymosan in the brain. The RTL9 protein is localized in the microglial lysosomes where incorporated zymosan is digested."*

The functional requirement of the retroviral Gag domain was established by a **structure–function loss-of-function experiment**: in *Rtl9* knockout mice engineered to express a truncated **RTL9ΔC** protein lacking the C-terminal retroviral GAG-like region, *"the zymosan degrading activity was lost."* This directly links the antifungal activity to the inherited retroviral capsid-like module rather than to the poorly characterized N-terminus.

An independent 2026 review (**Irie et al.**, *Virus-Derived Domesticated Genes in Microglia and Resident Macrophages*; PMID: 42511665) corroborates this, stating that *"SIRH3/RTL6, SIRH8/RTL5, and SIRH10/RTL9 mediate innate immune responses against bacterial-, viral-, and fungal-derived pathogen-associated molecular patterns, respectively."* RTL9 (= SIRH10) is thus consistently assigned the **antifungal** role.

Localization: microglial **lysosomes** (intracellular). **Substrate/target:** fungal cell-wall PAMP (zymosan / β-glucan). **Essential module:** C-terminal retroviral Gag/capsid-like domain.

Finding 2 — RTL9 is a domesticated Ty3/Gypsy (sushi-ichi) retrotransposon Gag-derived gene

RTL9 belongs to a family of **eleven** eutherian **RTL/SIRH** genes that were created by the **domestication (exaptation)** of a retroviral/retrotransposon Gag gene. As Kaneko-Ishino & Ishino, 2023 (*Retrovirus-Derived RTL/SIRH..., Biomolecules*; PMID: 37892118) state: *"Eutherians have 11 retrotransposon Gag-like (RTL)/sushi-ichi retrotransposon homolog (SIRH) genes presumably derived from a certain retrovirus."* The ancestral element is the **sushi-ichi LTR-retrotransposon**, a member of the **Metaviridae (Ty3/Gypsy)** superfamily first described in pufferfish. RTL9 is equivalently named **SIRH10** and **Mart9** in the Mart (Mammalian retrotransposon transcript) nomenclature.

The molecular signature of this retroviral ancestry is retained in UniProt Q8NET4 as the **PEG10-RTL1_C-capsid domain** (Pfam **PF30901** / InterPro **IPR063300**) — a Gag/capsid-like fold shared with the better-known domesticated genes **PEG10** and **RTL1/PEG11**. The persistence of

this domain over the ~160 million years of eutherian evolution, together with the functional data above, indicates that RTL9 was **repurposed** from a mobile-element gene into a host gene with a defined physiological role.

Finding 3 — RTL9 is the antifungal member of a three-gene microglial innate-immune module

A central organizing concept is that domesticated retroviral Gag genes built a **novel innate-immune system in the eutherian brain**. Kaneko-Ishino & Ishino, 2023 (PMID: 37892118) describe that *"at least three RTL/SIRH genes, RTL5/SIRH8, RTL6/SIRH3, and RTL9/SIRH10, play important roles in combating a variety of pathogens, namely viruses, bacteria, and fungi, respectively."* Each family member recognizes a **distinct class of PAMP**:

RTL / SIRH gene	Pathogen class	PAMP target	Site of action
RTL5 / SIRH8	Viruses	Double-stranded RNA	Extracellular (secreted granules)
RTL6 / SIRH3	Bacteria	Lipopolysaccharide (LPS)	Extracellular (secreted granules)
RTL9 / SIRH10	**Fungi**	**Zymosan / β-glucan**	**Intracellular (microglial lysosome)**

This division of labor means RTL9 occupies the **antifungal arm** of a dedicated, eutherian-specific defense system in the central nervous system. The module is paralogous — the three genes derive from the same ancestral Gag gene but have sub-functionalized to cover different microbial threats.

Finding 4 — Human gene/protein architecture: X-linked, 1388 aa, single C-terminal Gag domain, testis-enriched expression

According to NCBI Gene (ID **57529**), human RTL9 is a **single-copy gene on the X chromosome at Xq23** (GRCh38 chrX:110,358,847–110,456,333), with aliases **MART9** / **RGAG1** / **SIRH10**. UniProt **Q8NET4** describes a **1388-residue (~144 kDa)** protein. Domain mapping by InterPro/Pfam places the **only recognizable structured module** — the PEG10/RTL1 C-terminal capsid-like (Gag) domain (**IPR063300** / **PF30901**) — at approximately **residues 1239–1305**. The large **N-**

terminal region (residues 1–1238) contains multiple predicted intrinsically disordered segments and **no other recognized domains**, and UniProt provides **no curated function or subcellular-location annotation** for the human protein.

Expression data from the **Human Protein Atlas** indicate the RTL9 transcript is **"tissue enriched" in testis** (~7.5 nTPM) and **"cell-type enriched,"** with no experimentally determined subcellular localization in that resource. The testis enrichment in humans contrasts with the functionally characterized microglial/brain role in mice, a point revisited under Limitations. Consistent with low somatic expression in some tissues, an earlier mouse study (Kaneko-Ishino lab placentogenesis survey; PMID: 25888968) found **Rgag1/Mart9 expression undetectable in the placenta**, indicating RTL9 has no placental role and is spatially restricted — compatible with a specialized, tissue-specific (brain microglia / testis) function rather than a housekeeping one.

Finding 5 — Mechanistic model: RTL/SIRH Gag proteins are PAMP-binding/clearance effectors; RTL9 acts intracellularly at the lysosome

The mechanistic framework for the whole family comes from **Irie et al., 2022** (*Retrovirus-derived RTL5 and RTL6 genes are novel constituents of the innate immune system in the eutherian brain*, *Development*; PMID: 36162816). This study showed that **RTL5 and RTL6 are microglial proteins** that appear as **extracellular dots/granules** in the CNS and *"display a rapid response to pathogens such as lipopolysaccharide (LPS), double-stranded (ds) RNA analog and non-methylated CpG DNA, acting both cooperatively and/or independently."* Knockout experiments confirmed RTL6 acts against LPS (bacteria) and RTL5 against dsRNA (viruses). The paper explicitly provides *"the first demonstration that retrovirus-derived genes play a role in the eutherian innate immune system."*

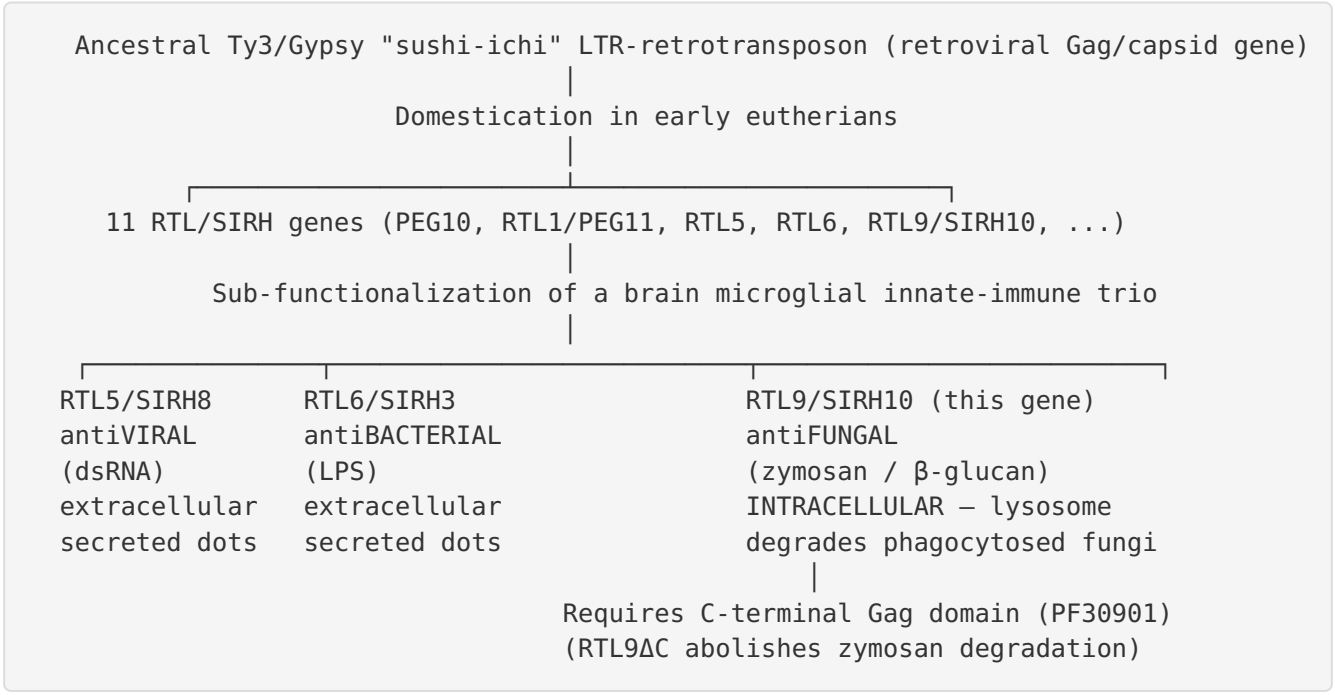
The unifying mechanism is therefore that **domesticated retroviral Gag proteins directly bind and neutralize or clear microbial molecular patterns**. RTL9 fits this framework but is mechanistically **distinctive**: rather than acting extracellularly, it operates **intracellularly within the microglial lysosome**, where it degrades phagocytosed fungal zymosan, with its C-terminal Gag domain being indispensable (PMID: 37834332). The retroviral capsid-like fold — evolutionarily built to package and interact with nucleic acids and to oligomerize — is plausibly repurposed here to **bind fungal cell-wall polysaccharide and promote its lysosomal breakdown**, although the precise biochemistry (direct β-glucan binding vs. recruitment of hydrolases) remains to be resolved.

Finding 6 — Human and mouse RTL9 are conserved orthologs with a highly conserved Gag domain

To assess whether the mouse-derived functional conclusions transfer to humans, the orthologs were compared directly. Human RTL9 (**Q8NET4**, 1388 aa) and mouse Rtl9 (**Q32KG4**, 1339 aa) are **single orthologs of comparable length**, and both retain the Pfam **PEG10/RTL1 C-terminal capsid-like Gag domain (PF30901)** — presence in the mouse protein was confirmed via InterPro. A **Needleman–Wunsch global alignment** of the C-terminal Gag-domain region yielded **~78% amino-acid identity (112/143 aligned positions)**. For a retroelement-derived protein, this level of conservation is high and indicates **strong purifying (negative) selection** acting specifically on the functional Gag module. This evolutionary signature strengthens the inference that the **antifungal function demonstrated in mouse microglia is conserved in the human protein**.

Mechanistic Model / Interpretation

The findings integrate into a coherent evolutionary-and-functional narrative:



What RTL9 does (primary function): It is an **innate-immune effector** that mediates the **lysosomal degradation of fungal cell-wall material (zymosan/β-glucan)** inside brain microglia. It is best classified as a **domesticated retroviral Gag-derived clearance/effector protein**, not an enzyme with a defined EC reaction, a transporter, or a signaling receptor.

Where it acts (localization): Intracellularly, in the **lysosomes of microglia** (brain-resident macrophages). This contrasts with its sister genes RTL5/RTL6, which act extracellularly as secreted granules — RTL9 is the **intracellular** member of the trio.

Pathway context: RTL9 functions within **innate antifungal immunity / phagocytic clearance** in the CNS. The phagocytosis → lysosomal trafficking → degradation axis is the operative pathway. It is specifically the **fungal-PAMP-handling** component, complementing (but distinct from) classical fungal-sensing pathways such as Dectin-1/β-glucan signaling. The evidence places RTL9 at the **effector/degradation** step rather than the receptor/signaling step.

Why the Gag domain matters: The retroviral capsid-like fold is the **essential functional unit** (RTL9ΔC loses activity). Evolutionarily, capsid proteins oligomerize and bind macromolecules; this biochemical capacity appears to have been exapted for engaging fungal polysaccharide and/or organizing its lysosomal degradation.

Evidence Base

PMID	Study (short)	Relevance to RTL9 function	Strength
37834332	Shiura et al. 2023, *IJMS* — RTL9 gene	**Primary functional evidence.** RTL9-mCherry knock-in: lysosomal localization in microglia; degrades zymosan; RTL9ΔC loses activity → Gag domain essential	**High** (direct, in vivo, loss-of-function)
37892118	Kaneko-Ishino & Ishino 2023, *Biomolecules* — RTL/SIRH review	Establishes RTL9 = SIRH10 = 1 of 11 domesticated retroviral Gag genes; antifungal member of microglial trio	High (authoritative review)
42511665	Irie et al. 2026 review — virus-derived genes in microglia	Independent confirmation: SIRH10/RTL9 mediates innate immunity against fungal PAMPs	High (independent review)
36162816	Irie et al. 2022, *Development* — RTL5 & RTL6	**Mechanistic framework** for the family: microglial Gag proteins directly respond to PAMPs; first demonstration of retrovirus-derived innate immunity in eutherian brain	High (direct, in vivo)
25888968	Mart-gene placentogenesis survey	Rgag1/Mart9 undetectable in placenta → RTL9 is spatially restricted, no placental role	Moderate (expression)

Incidental / non-functional literature (RGAG1 as a genomic locus — not function): RGAG1-BCOR gene fusions in sarcomas (PMID: 34331570, PMID: 32156473); Xq22.3-q23 deletion syndromes including RGAG1 as a bystander (PMID: 23520119); testicular germ-cell-tumor DNA-methylation dysregulation of RGAG1 (PMID: 20051947); and an RGAG1 association signal with LDL cholesterol in UK10K (PMID: 26367797). These report the locus but **do not characterize the protein's biochemical function** and should not be over-interpreted as functional evidence.

Limitations and Knowledge Gaps

1. **Species gap between function and human expression.** The direct functional evidence (lysosomal antifungal activity in microglia) comes from **mouse** models (PMID: 37834332). In humans, the transcript is reported as **testis-enriched** (Human Protein Atlas), and no human in-brain functional study exists. The strong Gag-domain conservation (~78% identity) supports — but does not prove — transfer of function to humans. Human microglial expression and activity need direct confirmation.
2. **No defined enzymatic mechanism.** It remains unresolved whether RTL9 itself catalyzes degradation, directly binds β-glucan via its capsid fold, or acts as a scaffold/adapter that recruits lysosomal hydrolases. The term "degrades zymosan" describes the phenotype, not a proven catalytic reaction; no EC number or substrate kinetics are available.
3. **The N-terminus (res 1–1238) is uncharacterized.** The vast majority of the protein lies outside the Gag domain and has no annotated function or structure. Whether it contributes to targeting, oligomerization, or substrate engagement is unknown.
4. **No experimental human subcellular localization.** Lysosomal localization is established in mouse microglia; the human protein's localization has not been experimentally determined.
5. **Sparse primary literature.** Functional understanding rests heavily on a small number of papers from a single research group. Independent replication would strengthen confidence.
6. **Clinical significance unclear.** Genomic associations (BCOR fusions, Xq deletions, TGCT methylation, LDL GWAS) are not mechanistically linked to the antifungal function and may be incidental to the locus.

Proposed Follow-up Experiments / Actions

1. **Confirm human microglial expression and localization.** Immunostain human brain microglia (and iPSC-derived microglia) for RTL9; use fractionation/confocal co-localization with lysosomal markers (LAMP1/2) to test whether the mouse lysosomal localization is conserved.

2. **Reconstitute the antifungal activity in human cells.** Express human RTL9 and RTL9ΔC in human macrophage/microglial lines, challenge with fluorescent zymosan/β-glucan, and quantify lysosomal degradation to confirm the Gag domain requirement in the human ortholog.
3. **Define the biochemical mechanism.** Test direct binding of purified RTL9 Gag domain to β-1,3-glucan/zymosan (pulldown, ITC, SPR); determine whether RTL9 possesses intrinsic hydrolytic activity or acts as a scaffold recruiting glucanases/cathepsins (proximity proteomics / BioID in lysosomes).
4. **Structural characterization.** Solve or predict (AlphaFold) the structure of the RTL9 C-terminal Gag domain and compare to PEG10/RTL1 capsids; probe whether it oligomerizes and whether a putative polysaccharide-binding surface exists.
5. **Dissect the N-terminal region.** Use truncation/domain-swap constructs and disorder-aware analysis to assign any role (trafficking, oligomerization) to residues 1–1238.
6. **In vivo infection models.** Challenge Rtl9-knockout vs. wild-type mice with live neurotropic fungi (e.g., *Candida*, *Cryptococcus*) to test whether RTL9 confers resistance to genuine CNS fungal infection, beyond model zymosan.
7. **Reconcile testis expression.** Investigate whether RTL9 has a second, germline-specific role in testis (e.g., in spermatogenesis or control of endogenous retroelements), which could explain the human expression pattern.

Conclusion

RTL9/RGAG1/SIRH10 (Q8NET4) is a **domesticated Ty3/Gypsy sushi-ichi retrotransposon Gag-derived gene**, X-linked (Xq23), encoding a 1388-aa protein whose sole functional module is a **C-terminal retroviral Gag/capsid-like domain** (PF30901). Its **primary function is innate antifungal immunity**: in brain **microglia** it localizes to **lysosomes** and mediates the **degradation of phagocytosed fungal cell-wall material (zymosan/β-glucan)**, an activity that **requires the Gag domain** (demonstrated by RTL9-mCherry knock-in and RTL9ΔC loss-of-function in mice). It is the **antifungal member of a eutherian-specific microglial innate-immune trio** (RTL5→viruses, RTL6→bacteria, RTL9→fungi) and is distinctive in acting **intracellularly at the lysosome** rather than as an extracellular granule, an enzyme with a

defined EC reaction, a transporter, or a signaling receptor. Strong human–mouse Gag-domain conservation (~78% identity) supports conservation of this function in humans, though direct human functional validation remains an important gap.
